

# Jonathan Cressman

## Staff / Principal Engineer, Embedded Systems

**Summary** 15+ years designing, developing and shipping resource constrained devices. Specialized in protocol stacks, Zigbee, IoT, secure machine to machine communications. Proven record creating robust, simple to follow modular code.

**Achievements** Contributor to the Zigbee, Thread and Matter IoT protocols. Primary author of multiple protocol features.

Cleaned up technical debt, increased team efficiency, solved concurrency issues

**Philosophy** I believe in ownership of design and code, looking at the big picture and the customers experience, keeping things simple and at the end of the day helping sales profitably sell our products

## Experience

**Universal Electronics** Dec 2024 – Dec 2025 (Full Time)  
**Sr Systems Architect Zigbee** Santa Ana, California

- Reverse-engineered and fully documented undocumented legacy Zigbee stack, restoring long-term maintainability and eliminating single-point knowledge risk
- Authored formal functional requirements and lifecycle process adopted by 15-engineer team
- Created layer-isolation and modular design and command line interface which greatly increased team efficiency
- Resolved Zigbee interoperability with diverse vendors
- Wrote UART and SPI drivers, interrupt handlers, command line interface
- Improved stability, including reducing crashes by 90% and increasing diagnostics for the remaining crashes

**Corellium** March 2023 – Feb 2024 (Contract)  
**Hardware modelling** Remote/Florida

- Modelling Silabs efr32mg24 system on a chip in software
- emulated UARTs, SPI, radio, DMA, crypto modules
- Reverse engineered how the processor works
- Created a virtual 802.15.4 network

**Swidget** Sept 2022 - March 2023 (Contract)  
**Principal IoT Developer** Dec 2020 - April 2022 (Full Time)  
Ottawa, Canada

- Created a single firmware build that can handle nearly 400 permutations of hardware allowing a simpler manufacturing process
- Scoped requirements, wrote code, tested and integrated with other vendors' devices
- Designed code to be modular with clean layers of abstraction allowing code to be reused on two other projects
- Quickly wrote multiple drivers for replacement board components due to component and chip shortages
- Wrote SPI, UART and I2C interfaces with hardware abstraction to diverse board components such that the drivers could be used on other Swidget devices

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## Skills

**Languages** C, C++, C#, Assembly, PHP, Rust, Python

**Protocols** Zigbee, 802.15.4, Thread, Matter, 6LoWPan, CoAP, TCP, UDP, IP, SPI, I2C, JTAG, DHCP, MQTT, UART, REST, HTTP, TLS

**Sensors** Air Quality, motion, temperature, humidity, luminosity, occupancy, pressure, vibration, current

**Cryptography** Post quantum cryptography, elliptic curves, signatures, FIPS 140

**Tools** Logic analyzers, Windows, Linux, Git, SVN, Eclipse, multi meter

**Certifications** ISO 17025, Common Criteria Assurance Level 4

## Interests

Security, post quantum computing, cryptocurrencies and decentralized systems

live action role playing, hiking

## Projects

### Zigbee ZTT Test Automation contributor

- Created test scripts for testing Zigbee devices
- Tested and debugged the Zigbee alliance test tool
- Documented how to use the tool and insights to the tool's working

### Cryptocurrency Review

- Reviewed technical papers and provided expertise for the privacy cryptocurrencies Monero and Zcash

## Education

### Bachelor of Math

### University of Waterloo

- Specialization in Pure Mathematics and Combinatorics and Optimization

## Experience Continued

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### **Alarm.com**

April – Nov 2022 (Contract)

#### **Embedded Engineer - DSC Alarm Panel Integration**

Remote

- Rewrote entire application code resulting in reduced image size (145KB to 45KB) allowing OTA upgrades
- Added structure and modularization to the code to allow the code to be reused and more easily maintained
- Documented the DSC Panel protocol and alarm.com's functionality
- Cleaned up 10+ years of technical debt
- Rewrote the interrupt and signal handling code to eliminate race conditions and edge cases

### **Machine to machine communications,**

Jan 2003 – June 2020 (Various Contracts)

#### **Wireless IoT Protocols and Cryptocurrencies**

Ottawa, Canada/Remote

- Reverse engineered, thoroughly documented, and cleaned up code for various projects
- Analyzed viability of the Thread wireless mesh networking protocol for clients' IoT devices
- Verified proofs of security and reviewed academic papers for cryptocurrencies
- Validated and tested cryptographic systems with short deadlines and no tolerance for errors

### **Xylem Inc.**

Jan 2017 – June 2020 (Contract)

#### **Principal IoT Developer**

North Carolina/Remote

- Worked on the leading edge of Zigbee, first to implement new features
- Anticipated customer needs early on, implemented key additional flexibility features ahead of 5+ year utility upgrade cycle
- Co-authored Zigbee electric vehicle supply management cluster
- Doubled Zigbee network joining test coverage, and rewrote 75% of legacy tests leading to 50% faster test execution time
- Led Zigbee spec modification solutions to solve multi-vendor legacy interoperability issues

### **Entrust Datacard**

2015 – 2016 (Full Time)

#### **Security Assurance**

Ottawa, Canada

- Tracked security vulnerabilities across all Entrust products
- Evaluated and prioritized vulnerabilities in both Entrust code and in third party tools incorporated into Entrust products
- Evaluated new technologies and developments in cryptography, specializing in post quantum computing

### **Energate Inc.**

2008 – 2015 (Full Time)

#### **Embedded IoT Engineer**

Ottawa, Canada

- Implemented Zigbee protocol firmware code for gateways, thermostats and load control devices
- Advocated for simplified standards at Zigbee committee meetings, negotiated with stakeholders to improve the specifications
- Communicated with several companies to facilitate interoperability between Energate devices and their devices
- Helped our partners troubleshoot network communication issues
- Avoided detrimental issues in early spec iterations by leading development of both client and server implementations

### **IPeak Networks/LiveQOS**

2007 – 2008 (Full Time)

#### **System Architect**

Ottawa, Canada

- Designed and engineered a system that became the company's main product for the next 10+ years
- Designed a protocol for hijacking an existing TCP/UDP connection to run proprietary forward error correcting algorithm over the connection which was later granted a patent
- Implemented code for initial product design, ultimately implemented test plan and fixed all bugs
- First engineer hired; company later grew to over 100 employees

### **Pre 2007**

- Maintained the BSD based TCP/IP network stack for VxWorks used in devices with hard real time communications constraints
- Wrote assembler code for a 16 core 128 thread network processor
- Increased packet throughput 11x, increased traffic analysis scope, assisted sales, marketing and hardware design
- Implemented a high level scripted language on a network processor
- Evaluated security products to FIPS 140, Common Criteria and other standards
- Taught IBM internal cryptography courses to other engineers
- Devised both physical and logical attacks against secure devices - required learning new products, reading source code, PCB layouts, writing specialized code